

Question 5**PAC - Problem Analysis Chart****Given Data****Required Results**

- | | |
|-----------------------------------|-------------------------------------|
| Vehicle count N | Accepted / rejected vehicles. |
| vehicle type: C / B / V | Cars, bikes & Vans parked |
| Category: F / S / G | Zone occupancy & remaining capacity |
| Permit: Y / N | Zone with highest occupancy |
| Emergency: Y / N | Campus full status |
| Zone capacities: A=20, B=40, C=15 | Parking summary |

Process Required Alternation Solution

- | | |
|---|---|
| Input vehicle type, category, permit & emergency status | Use separate functions for validation, eligibility and zone assignment. |
| Validate all inputs using loops | Use a single structured program decision system in the main |
| check eligibility and available capacity | |
| Assign preferred / alternative zone | |
| Update occupancy & counters | |
| Repeat for all vehicles. | |

IPO Chart

Input	Process	Output
- Number of vehicles: N	Validate vehicle type, category and permit	Accepted/Rejected status
- Vehicle type: C/B/V	Check emergency status if permitted N	Assigned parking zone
- User category: F/S/G	Determine eligibility and suitable zone	Rejection reason
- Parking permit: Y/N	Check available capacity	Remaining zone capacity
- Emergency status: Y/N	Assign vehicle and update occupancy	Total processed vehicles
- Zone capacities: A=20, B=40, C=15	Update accepted/Rejected and vehicle count	Accepted/Rejected totals
-	Report for all vehicles using loop	Cars, bikes, vans parked
-	Find highest occupied zone	Final occupancy of A, B and C
-	Check whether all zones are full.	Highest occupancy zone and campus full status

①

Date: _____

Pseudocode

Begin

Declare variables for vehicle type, user category, parking permit, emergency status, number of vehicles, zone, occupancies, accepted and rejected vehicles and number of cars, bikes and vans.

Set $i=0$, OccupiedA=0, OccupiedB=0, occupiedC=0, accepted=0, rejected=0, carCount=0, bikeCount=0 and vanCount=0

Input n

while ($i < N$)

$i \leftarrow i + 1$

Display vehicle number, i

Declare a, b, c, d as integer and set their values to 0

while a=0

Input type

If type is C, B, or V, c, b, or v
set a=1

else

Display "invalid vehicle type".

while b=0

Input category.

If category is F, S, G, f, s or g
set b=1

else

Display "invalid category tag again".

②

Date:

while c=0

Input permit

If permit is Y or N or n or y

set c=1

else

Display "Invalid permit value".

If permit is N or n

while d=0

‡ Input emergency vehicle status

If emergency is Y, y, N or n

Set d=1

else

Display "Invalid vehicle".

If category is F or f

If Zone A has space

• If vehicle is car

If permit is Y or y

OccupiedA ++

carCount ++

accepted ++

~~Else if typ~~

elseif vehicle is bike

If permit is Y or y

OccupiedA ++

BikeCount ++

accepted ++

elseif vehicle is van

If permit is Y or y

OccupiedC 2++

vanCount ++

accepted ++

else

‡ Increase rejected by 1

③

Date:

Else if category is S or s
if Zone B has space

If type is car

If permit is Y or y
Occupied B ++
Car Count ++
accepted ++

elseif type is bike

If permit is Y or y
Occupied B ++
Bike count ++
accepted ++

else if type is van

If permit is Y or y
Occupied B ++
van count ++
accepted ++

else

Rejected ++

Else if category is G or g
if Zone C has space

if type is car

if permit is Y or y
Occupied C ++
Car Count ++
accepted ++

elseif type is bike

if permit is Y or y
Occupied C ++
Bike Count ++
accepted ++

else if type is van

if permit is Y or y

(4)

Date:

Occupied++

vanCount++

accepted++

• else

rejected++

end while

Display total vehicles processed, n

Display total accepted vehicles, accepted

Display total rejected vehicles, rejected.

Display number of successfully parked cars, bikes, and vans.

Display occupied and remaining spaces of Zone A

Display occupied and remaining spaces of Zone B

Display occupied and remaining spaces of zone C

If occupied A is greater than or equal to both occupied B and occupied C

Display "Zone A has highest occupancy"

else if occupied B $\geq$ occupied A and occupied C

Display "Zone B has highest occupancy"

else

Display "Zone C has highest occupancy"

if zone A, Zone B and Zone C are fully occupied.

Display Entire campus parking facility is Full

else

Display Entire campus parking facility is Not Full

End.

Algorithm

1. Start
2. Initialize zone occupancies, accepted/rejected counts, and vehicle counters to zero
3. Input the number ~~of vehicle~~ of vehicle.
4. Repeat following steps for each vehicle.
 - Input and validate vehicle type, user category and parking permit.
 - if the permit is invalid, input and validate emergency status.
 - Check the user's category and select the appropriate parking zone
 - Check whether the selected zone has enough available space.
 - if the vehicle has a valid permit or is an emergency vehicle and space is available, accept the vehicle and update the required counters.
 - Otherwise reject the vehicle.
5. Display the total processed, accepted and rejected vehicles.
6. Display the number of cars, bikes and vans successfully parked.
7. Display occupied and remaining spaces for all three zones.
8. Determine and display the zones ^{that} are full and display the result.
9. Stop.